

Lesson Text

Unit I: Foundations

Module 1: What This Data Is (and Isn't)

What This Data Is (and Isn't): Situating the Minnesota Felony Sentencing Dataset

Every analytic task that follows be it filtering, summarizing, comparing groups, presupposes an answer to a fundamental question: what is the dataset students will be exploring, and what can it honestly tell them? That question is not a formality. Numbers about crime are among the most misread statistics in public life. In 23 of 27 national surveys conducted since 1993, a majority of Americans reported believing that crime was rising, even across long stretches in which recorded crime was falling (Gramlich, 2024). A statistic is only as trustworthy as one's understanding of where it came from and what it leaves out (Best, 2001). This chapter therefore traces the Minnesota felony sentencing dataset back to its origin—who created it, why, and at what point in the life of a crime it was recorded—so that a reader can state, in plain terms, which questions the roughly 294,000 felony sentencing records covering 2001 through 2019 can answer and, just as importantly, which they cannot.

Administrative Data: Built to Run a System, Not to Answer a Question

The dataset is administrative data: information generated as a by-product of running an organization rather than collected to answer a research question (Connelly et al., 2016). Courts and the Minnesota Sentencing Guidelines Commission record sentences because the state must document and monitor how people are punished, not because a researcher designed a study. That

origin is at once a strength and a limitation. Administrative records cover nearly every case, span decades, and are not subject to survey sampling error. Yet because the data was shaped by the needs of the system that produced it, using it for research requires reconstructing how each field was defined and recorded; the categories, rules, and exceptions were chosen by administrators, not by the analyst (Connelly et al., 2016). A useful discipline, then, is to treat every column as an administrative decision first and a measurement second, and to ask why the system would record a value in a particular way. That habit forestalls a great many misreadings.

The Source: The Minnesota Sentencing Guidelines Commission

In 1978 the Minnesota Legislature created the Minnesota Sentencing Guidelines Commission, and in 1980 Minnesota became the first state to sentence felonies under presumptive guidelines developed by such a commission (Frase, 2005; Minnesota Sentencing Guidelines Commission, n.d.). The guidelines derive a presumptive sentence for each case from two facts—the severity of the offense and the offender's criminal-history score—a structure intended to make punishment more uniform and principled (Frase, 2013). To monitor whether courts follow the guidelines, the Commission maintains a record of every person convicted of a felony in adult court and sentenced under them, assembled from probation officers' sentencing worksheets and court data (Minnesota Sentencing Guidelines Commission, 2024). The present dataset is a slice of that monitoring system. For scale, annual felony case volumes have ranged from roughly 5,500 in 1981 to a peak of 18,288 in 2017 (Minnesota Sentencing Guidelines Commission, 2023). This provenance matters for interpretation: the data exists to answer the Commission's question “are sentences following the guidelines?” and was never designed to measure how much crime Minnesota experiences, or why. Keeping that original purpose in view keeps one's conclusions honest.

The Criminal-Justice Funnel

A sentence is the last event in a long sequence. For a felony sentence to exist, a crime must have been committed, noticed, reported, recorded by police, and investigated; an arrest must have been made, charges filed, and guilt established; and a felony-level sentence must have been imposed. Criminologists picture this progression as a funnel: many events enter at the top, and only a fraction reach the bottom. The conception of criminal justice as a connected system, with cases dropping out at each stage, was articulated by the President's Commission on Law Enforcement and Administration of Justice (1967). The dataset sits at the very bottom of that funnel; it contains only cases that survived every prior stage and ended in a felony sentence. It therefore says nothing directly about crimes no one reported, reported crimes that produced no arrest, arrests that produced no charge or conviction, or offenses resolved as misdemeanors or dismissed. Each of these is a stage of the funnel the data never observes.

The Dark Figure of Crime

Criminologists call the crime that never appears in official records the dark figure of crime: occurrences that are treated as crime but registered in no agency's statistics (Biderman & Reiss, 1967). The dark figure is not small. In the National Crime Victimization Survey, only about 42% of violent victimizations were reported to police in 2022, meaning nearly six in ten went unreported (Morgan & Thompson, 2023). Nor is unreported crime a random sample of all crime; it skews toward less serious incidents, smaller losses, and less injury (Skogan, 1977). The crimes that do climb the funnel toward a felony sentence are thus systematically different from those that do not. Before a single row of the dataset was recorded, in other words, most crime had already dropped out of view—and dropped out selectively. No filter applied after the fact can recover it.

Sellin's Dictum: Distance From the Act

In 1931 the criminologist Thorsten Sellin articulated a principle now known as Sellin's dictum: the farther a statistic is collected from the criminal act itself, measured in procedural steps, the less reliable it is as an index of crime (Sellin, 1931). Police reports sit closer to the event than arrests, arrests closer than convictions, and convictions closer than sentences. Sentences lie as far down the procedural chain as it is possible to go. By Sellin's rule, felony sentencing data is therefore an excellent record of one thing—what the system decided to do—and a poor proxy for another—how much crime actually occurred. This single idea reframes the entire enterprise. The roughly 294,000 records constitute a rich, reliable account of Minnesota's sentencing decisions; read as a map of crime itself, they are among the weakest measures available. Almost every error in interpreting the data stems from confusing the two.

How the Nation Measures Crime, and How Minnesota's Data Differs

Because no single source captures all crime, the United States relies on two national systems that sit at different points in the funnel. The FBI's Uniform Crime Reporting program, now transitioning to the more detailed National Incident-Based Reporting System, counts crimes recorded by police (Federal Bureau of Investigation, n.d.). The Bureau of Justice Statistics' National Crime Victimization Survey instead asks a national sample of households about crimes they experienced, including those never reported to police (Bureau of Justice Statistics, n.d.-b). The two disagree by design, because they measure different slices of the funnel. The Minnesota data is neither of these; it is a court and sentencing record that sits below both in the funnel. Consequently, before comparing a finding from this dataset to a national crime statistic, one must ask whether the two even measure the same stage. Usually they do not.

The Unit of Analysis: What One Row Represents

A dataset is only as clear as its unit of analysis—what a single row represents. Here, one row is one sentencing record, and the Commission's counting rules define it precisely: when a person is sentenced in the same county for several offenses within a 30-day window, those offenses are combined into a single record described by the most serious offense (Minnesota Sentencing Guidelines Commission, 2018). Two consequences follow, and both are easy to mistake. First, one row is not one person: an individual sentenced in 2004 and again in 2015 occupies two rows, because the data counts sentencing events rather than individuals—the same distinction federal court statistics draw between a case and a defendant (Bureau of Justice Statistics, n.d.-a). Second, one row is not one crime: secondary offenses within a record are collapsed into the most serious one, and, as the funnel shows, most crimes never produce a record at all. Every total in the dataset should therefore be read not as a count of people or of crimes but as a count of sentencing records. The full dataset comprises roughly 294,000 such records described across more than 100 administrative fields—each field a category the Commission chose to record and, by the same token, a reminder of the many things it did not.

What the Dataset Excludes

Combining the funnel with the counting rules makes plain what these records omit. They exclude unreported crime—the dark figure, which constitutes the majority of offenses (Biderman & Reiss, 1967). They exclude crimes that produced no felony conviction, including cases dropped, dismissed, acquitted, or resolved as misdemeanors. They exclude secondary offenses, which are folded into a record's most-serious charge (Minnesota Sentencing Guidelines Commission, 2018). They exclude most white-collar and regulatory harm, which rarely yields a felony sentence—an offense that by definition may carry a term of at least a year. And they exclude context about

people's lives, since the monitoring system does not maintain data on offenders' social or economic circumstances (Minnesota Sentencing Guidelines Commission, 2024). A final caution belongs here: this is an educational dataset, not a prediction engine. It describes sentences that were imposed; it is not a calculator for the sentence a future case should or will receive. The gap between the guideline-recommended sentence and the sentence actually pronounced is itself a subtle matter, taken up in a later chapter.

Why a Pattern Is Not a Verdict: Description Versus Explanation

A sentencing record is the end product of a long chain of human decisions—victims who report or do not, police who arrest or do not, prosecutors who charge or bargain, and judges who sentence. Every stage filters which cases arrive at the bottom (Klepper et al., 1983). Studying only sentenced cases therefore means studying a selected, non-random sample of everything that happened (Bushway et al., 2007). When two groups exhibit different average sentences, one has found a description of the data, not its cause; correlation, as the adage holds, does not establish causation. A landmark review of sentencing research concluded that even after accounting for the two strongest legal factors—offense severity and criminal history—most of the variation in sentences remains unexplained, and that observed group differences cannot by themselves be read as proof of discrimination (National Research Council, 1983). Descriptive differences are real and worth identifying; explaining why one exists is a separate and harder task. The proper reading of a cross-tabulation showing that offenses in one category receive longer average presumptive sentences than those in another is simply that a descriptive difference exists in the data, with the reasons behind it requiring further, careful analysis.

The Stakes: Why Careful Reading Matters

The insistence on caution has a public rationale: crime statistics are unusually easy to misuse, and the consequences play out in policy and opinion. Despite a long-term decline in recorded violent crime since the early 1990s—the rate has fallen by roughly half—large majorities of Americans have repeatedly told pollsters that crime is worsening (Gramlich, 2024). People reason confidently from numbers they have not examined, precisely the failure that data literacy is meant to prevent (Best, 2001). Three questions guard against it: Where in the funnel does this number sit? What is its unit of analysis? And is a pattern being described or explained?

Summary

The core claims of this chapter can be restated compactly. The Minnesota felony sentencing dataset is administrative data—a by-product of running the state's sentencing system rather than a study designed to measure crime. This is the distinction between found and made data: both involve deliberate thought, but for made data the research question shapes the collection, whereas a monitoring dataset like this one is structured around the needs of the system that produced it (Connelly et al., 2016). The data sits at the bottom of the criminal-justice funnel, containing only cases that survived every earlier stage and ended in a felony sentence (President's Commission on Law Enforcement and Administration of Justice, 1967). Most crime lies in the dark figure and never enters the funnel at all, and it is absent non-randomly (Biderman & Reiss, 1967; Skogan, 1977). By Sellin's dictum, this makes the data an excellent record of decisions and a weak proxy for crime (Sellin, 1931). One row is one sentencing record—not one person and not one crime (Minnesota Sentencing Guidelines Commission, 2018). And a pattern in the data is a description, not an explanation; causes demand separate, careful analysis (National Research Council, 1983). Carried into every subsequent chapter, three questions keep interpretation honest: where in the

funnel a number sits, what its unit of analysis is, and whether a pattern is being described or explained.

References

Best, J. (2001). *Damned lies and statistics: Untangling numbers from the media, politicians, and activists*. University of California Press.

Biderman, A. D., & Reiss, A. J., Jr. (1967). On exploring the "dark figure" of crime. *The Annals of the American Academy of Political and Social Science*, 374(1), 1–15.

<https://doi.org/10.1177/000271626737400102>

Bureau of Justice Statistics. (n.d.-a). Federal criminal case processing statistics (FCCPS) tool: Supplementary information. U.S. Department of Justice. <https://bjs.ojp.gov/federal-criminal-case-processing-statistics-fccps-tool-supplementary-info>

Bureau of Justice Statistics. (n.d.-b). National Crime Victimization Survey (NCVS). U.S. Department of Justice. <https://bjs.ojp.gov/data-collection/ncvs>

Bushway, S., Johnson, B. D., & Slocum, L. A. (2007). Is the magic still there? The use of the Heckman two-step correction for selection bias in criminology. *Journal of Quantitative Criminology*, 23(2), 151–178. <https://doi.org/10.1007/s10940-007-9024-4>

Connelly, R., Playford, C. J., Gayle, V., & Dibben, C. (2016). The role of administrative data in the big data revolution in social science research. *Social Science Research*, 59, 1–12. <https://doi.org/10.1016/j.ssresearch.2016.04.015>

Federal Bureau of Investigation. (n.d.). National Incident-Based Reporting System (NIBRS). U.S. Department of Justice. <https://www.fbi.gov/how-we-can-help-you/more-fbi-services-and-information/ucr/nibrs>

Frase, R. S. (2005). Sentencing guidelines in Minnesota, 1978–2003. *Crime and Justice*, 32, 131–219. <https://doi.org/10.1086/655354>

Frase, R. S. (2013). *Just sentencing: Principles and procedures for a workable system*. Oxford University Press.

Gramlich, J. (2024, April 24). What the data says about crime in the U.S. Pew Research Center. <https://www.pewresearch.org/short-reads/2024/04/24/what-the-data-says-about-crime-in-the-us/>

Klepper, S., Nagin, D., & Tierney, L. J. (1983). Discrimination in the criminal justice system: A critical appraisal of the literature. In A. Blumstein, J. Cohen, S. E. Martin, & M. H. Tonry (Eds.), *Research on sentencing: The search for reform* (Vol. 2, pp. 55–128). National Academy Press.

Minnesota Sentencing Guidelines Commission. (n.d.). About MSGC. <https://mn.gov/sentencing-guidelines/about/>

Minnesota Sentencing Guidelines Commission. (2018). *Sentencing practices: Annual summary statistics for felony cases sentenced in 2017*. <https://mn.gov/msgc-stat/documents/reports/2017/UpdatedMSGC2017AnnualSummaryStatistics.pdf>

Minnesota Sentencing Guidelines Commission. (2023). *2023 sentencing practices report: Summary statistics for felony cases*. https://mn.gov/sentencing-guidelines/assets/2023_MSGC_Annual_Summary_Statistics_tcm30-680133.pdf

Minnesota Sentencing Guidelines Commission. (2024). *Report to the legislature*. <https://www.lrl.mn.gov/docs/2024/mandated/240055.pdf>

Morgan, R. E., & Thompson, A. (2023). *Criminal victimization, 2022* (NCJ 307089). Bureau of Justice Statistics. <https://bjs.ojp.gov/library/publications/criminal-victimization-2022>

National Research Council. (1983). *Research on sentencing: The search for reform* (A. Blumstein, J. Cohen, S. E. Martin, & M. H. Tonry, Eds.). National Academy Press. <https://doi.org/10.17226/19436>

President's Commission on Law Enforcement and Administration of Justice. (1967). *The challenge of crime in a free society*. U.S. Government Printing Office.

Sellin, T. (1931). The basis of a crime index. *Journal of Criminal Law and Criminology*, 22(3), 335–356.

Skogan, W. G. (1977). Dimensions of the dark figure of unreported crime. *Crime & Delinquency*, 23(1), 41–50. <https://doi.org/10.1177/0011826X770230010041>

Module 2: Driving the Explorer

Driving the Explorer: Filtering, Summarizing, and Cross-Tabulating the Sentencing Data

This chapter is a working tour of the Minnesota Sentencing Explorer, the interface through which the felony sentencing dataset is examined throughout the course. It assumes no statistical background. Its aim is practical: to establish the five operations used by students, which include filtering, reading descriptive statistics, reverting, clearing, and cross-tabulating. Together, these are sufficient to build almost any view of the data. The dataset comprises roughly 294,000 Minnesota felony sentencing records from 2001 through 2019, and, as the previous chapter established, each row is one sentenced count not one person, and not one crime committed. Three columns recur in what follows: *sentyear*, the year the sentence was imposed; *time*, the presumptive sentence in months; and *moc1*, the first character of the Minnesota Offense Code, which names the top-level offense category.

The Starting Point: The Whole Dataset

With no filters applied, every row is in view. The most important quantity on screen is the count, usually written *N*: the number of rows currently included. Because every operation that follows is measured by its effect on *N*, locating that number is the first habit to build. On the full, unfiltered dataset, *N* is approximately 294,000.

Filtering

A filter keeps only the rows that match a condition and hides the rest. The Explorer offers the standard comparisons: equals (eq), does not equal (ne), greater than and greater-than-or-equal (gt, ge), and less than and less-than-or-equal (lt, le). Two properties matter. First, filtering never deletes data; it narrows what is being looked at, and the underlying records remain untouched. Second, filters chain: each new filter applies only to the rows that survived the previous one. The Explorer

preserves this chain as an ordered filter history, and the list shown on screen is that history.

Applying a single filter—for instance, `sentyear >= 2015`, which keeps only counts sentenced in 2015 or later—adds one entry to the history and reduces N below its full-dataset value.

Reading a Column Summary

For a numeric column such as time, the summary reports several descriptive statistics, of which two deserve immediate attention. The mean is the arithmetic average—every value summed and divided by N . The median is the middle value, with half the rows below it and half above. The two answer slightly different questions and are frequently not the same number. Sentencing data illustrates why: a small number of very long sentences can pull the mean upward while barely moving the median. The gap between them is itself informative—a matter a later chapter takes up—but for now it is enough to locate both in the summary and keep them distinct.

Chaining Filters and the Offense-Code Drill-Down

Because filters stack, adding a second filter applies it only to the rows that survived the first; the history grows by one entry, and N can only stay the same or shrink. The Minnesota Offense Code receives special handling. Rather than entering a full code, one drills down one character at a time. The first character, `moc1`, is the top-level offense category—`moc1 = A`, for example, is assault. A drill-down is a filter like any other and joins the same history. Applying `moc1 == A` on top of `sentyear >= 2015` produces a two-entry history, in the order the filters were applied, and a correspondingly smaller N : the second filter retains only the assault counts among the rows the year filter had already isolated.

Revert and Clear

Storing filters as an ordered history makes undoing precise. Revert removes the most recent entry, stepping back one filter. Clear empties the entire history, returning to the full dataset. Neither

operation loses anything: the data is untouched, and any view can be reconstructed by re-applying the same filters in the same order. Reverting the assault drill-down from the two-filter view, for instance, restores exactly the earlier state in which only the year filter remained, and N returns to precisely the value it held before the drill-down was added. That equality is a useful check that `revert` does what it claims.

Cross-Tabulation

A cross-tabulation splits the data by two grouping columns at once and reports, in each cell, the N , mean, median, and standard deviation of a third, numeric column. It answers questions of the form: how does presumptive sentence length differ across offense categories, year by year? Reading it is a matter of intersection—choosing a row (an offense category) and a column (a year) identifies the single cell describing only the sentences in that category and that year. A representative table might report the mean of time, with `sentyear` across the columns and `noc1` down the rows, so that the cell where assault meets 2018 gives the mean presumptive sentence for assault counts sentenced in that year.

The idea that demystifies the crosstab is that every cell is a filtered dataset one could have built by hand. The assault-in-2018 cell is exactly the view produced by the filter chain `sentyear == 2018` followed by `noc1 == A`; the mean shown in that cell and the mean in the corresponding column summary are the same number, computed from the same rows. A crosstab, in other words, is a compact array of filtered views, not a new kind of object.

One caution carries over from the previous chapter and belongs here from the outset: a difference between two cells is a description of the data, not an explanation of why that difference exists. Establishing that a gap is real and establishing why it exists are separate tasks, and the latter is the subject of a dedicated later chapter.

Summary: The Five Operations

The five operations, stated compactly: a filter narrows the current view and joins the history; the column summary reports descriptive statistics for any column of that view; revert steps the history back by one; clear empties it; and a cross-tabulation compares a numeric column across two groupings, where every cell is itself just a filtered view. Building a specific view is then a matter of composing these operations in order—clearing the history, filtering sentyear to equal 2018, and drilling moc1 into assault, for example, reproduces the assault-in-2018 cell from first principles. Everything in the chapters that follow is assembled from these same moves.

Module 3: What the Numbers Say

What the Numbers Say: Reading Descriptive Statistics

The operations of the previous chapter produce views of the data; this chapter concerns the numbers those views report. Every statistic considered here is a descriptive statistic: a summary of what is in the current view. Descriptive statistics report patterns; they do not explain why a pattern exists, and that line stays bright throughout. The Explorer computes five such summaries. For categorical columns, which hold labels, it reports value counts and percentages. For numeric columns it reports those plus the mean, median, and standard deviation. The columns that recur below are time, the presumptive sentence in months; moc1, moc2, and moc4, successive characters of the Minnesota Offense Code; sex, the sentenced person's sex as recorded by the court; history, the criminal-history score, where zero denotes no prior history; and tweight, the combined severity weight of additional felony counts at sentencing.

The Mode: The Most Common Category

For a categorical column, the natural summary is the mode—the single most frequent value. Reading it means reading three parts together: the value itself (an offense category), its count (how many rows hold that value), and its percentage (that count as a share of all rows in view). On the full, unfiltered dataset, the mode of moc1 is drug offenses, the largest single block of felony sentencing in Minnesota.

Counts Versus Percentages

A raw count answers *how many*; a percentage answers *how common*. These are different questions, and they can disagree. When one group is far larger than another, the larger group will out-count the smaller on nearly everything—even on characteristics that are actually rarer within it—so whenever groups of different sizes are compared, the percentage is usually the honest

measure. Sentence length by sex illustrates the trap directly. Among women, the "year and a day" sentence—recorded as 12.03 months, since 366 days edges past the one-year felony threshold—accounts for a substantial share of all sentences, even though women are a small minority of sentenced counts. Among men, the same 12.03-month value has a far larger raw count but a far smaller percentage. The pattern is well documented in criminal-justice data: women enter the system far less often, and when they do their sentences tend to run shorter. Reading that pattern correctly requires the percentage, which adjusts for how few women appear in the data at all; the raw count, dominated by the size of the male group, would mislead. Why the pattern exists is a genuinely hard question reserved for a later chapter; the present task is only to read it.

Two Kinds of Middle: Mean and Median

Numeric columns admit two measures of center. The mean sums every value and divides by N . The median sorts the values and takes the middle one, so that half the rows fall below it and half above. A minimal example fixes the difference. For the five numbers 2, 3, 4, 5, and 26, the median is 4—the middle value—while the mean is 8, dragged upward by the single large value. That example is, in miniature, the entire behavior of sentencing data.

Consider first a view in which the two agree. Filtering to assault (`moc1 == A`) and then to terroristic threats (`moc2 == 9`) isolates an offense that carries short expected sentences; the genuinely serious cases tend to be charged as something more severe, so the view holds mostly short sentences with few extreme values and no long tail. There the mean and median of time sit close together—the numeric signature of a roughly symmetric distribution.

Clearing the filters and reading time across all felonies produces the opposite. Most sentences are short—drug offenses, the largest block, typically run about a year—but the same view contains murder, where sentences extend to decades. Those rare, very large values form a long right tail,

and the mean consequently sits well above the median. This is the arithmetic of 2, 3, 4, 5, 26 at full scale: the tail pulls the mean while the median holds its position. As a general reading rule, a mean noticeably above the median indicates a long right tail—many short sentences alongside a few very long ones.

Why the Median Is the Usual "Typical," and When the Mean Still Matters

Returning to the example once more and replacing the 26 with a 6 drops the mean from 8 to 4 while leaving the median unchanged: one value moved, and only one statistic moved with it. The median is, in the statistical sense, robust—it resists extreme values—whereas the mean is not, since every value exerts a pull on it and a single decades-long sentence can shift the mean of an entire year of data. Because the long tail in felony sentencing is not an error to be cleaned away but a permanent feature of the system, researchers and careful journalists habitually report the median sentence as the "typical" one. Encountering a claim that "the average sentence for some offense is 40 months" should therefore prompt an immediate question: mean or median?

The mean is not thereby a mistake to be avoided; it answers a different question. Because the mean multiplied by N yields the total, it is the right instrument whenever total burden is what matters. A corrections department projecting prison capacity cares about the total months to be served, and a budget office estimating annual cost reasons the same way—both are mean questions. A reporter describing what happens to a typical defendant is asking a median question. The competence being built is not "median good, mean bad" but matching the statistic to the question—typical case or total burden—and noticing which one a given claim is quietly relying on.

Spread: The Standard Deviation

Two views can share a mean and still look nothing alike—one huddled tightly around it, the other scattered across a wide range. The standard deviation measures this, capturing, roughly, the

typical distance of a row from the mean. A large standard deviation means the values are widely scattered and the mean says little about any individual case; a small one means the values huddle close and the mean is a good guess for nearly every case. No formula is needed at this stage; like the mean–median gap, the standard deviation is read by comparison.

Both extremes have already appeared. Across the whole dataset, where sentences range from probation length to decades, the standard deviation of time is enormous—comparable in size to the mean itself. Within terroristic threats, where short sentences bunch tightly, it is small. The statistic and the column are identical in the two views; the only thing that differs is how alike the cases are.

[FIGURE PLACEHOLDER 1: distribution of time, full dataset—wide, right-skewed]

[FIGURE PLACEHOLDER 2: distribution of time, terroristic threats only—narrow, compact]

Two readings follow directly. When the standard deviation of time is nearly as large as its mean, the sentences are widely scattered and the mean is a poor guess for any individual case. When it falls close to zero, nearly every sentence sits near the mean and the cases are highly alike.

Where Spread Comes From: Variance as Composition

The deepest idea in this chapter concerns the origin of spread. Reading ‘tweight’ which is the combined severity weight of additional felony counts on the unfiltered dataset yields a large standard deviation: sentenced counts differ enormously in how much extra felony weight accompanies them. The question is whether that spread reflects erratic sentencing or a view that mixes very different kinds of cases together. Filtering to history == 0, people with no criminal history, answers it: the standard deviation of ‘tweight’ collapses to roughly a sixteenth of its former value. Nothing about the measurement changed; the composition of the view did. Within a group of

similar cases, the cases are, unsurprisingly, similar. Much of the spread visible in any large dataset is therefore not randomness but mixture. This is why filtering is more than a search tool—every filter poses a question of the form "how much of the pattern survives when I hold this constant?"—and that move is the heart of a later chapter.

The pieces combine into a single reading. A column reporting a mean of 45 months, a median of 21 months, and a standard deviation of 60 months describes a distribution in which most sentences are fairly short, a few are much longer (the mean sitting above the median marks the long right tail), and the values overall are spread very widely (the large standard deviation).

The Final Rule: Check *N* First

One rule governs all of the above: every statistic is only as trustworthy as the *N* beneath it. The point is best made by an extreme case. Filtering to weapons offenses (`moc1 == W`) and then to the fireworks sub-code (`moc4 == 8`) returns every weapons-by-fireworks felony Minnesota sentenced in roughly two decades—exactly four cases, three for possession of illegal fireworks and one for altering serial numbers. The Explorer will still dutifully compute a mean, a median, and a standard deviation of time to two decimal places, but with *N* = 4 those are not statistics so much as four anecdotes in formal dress. Before trusting any summary, in other words, check *N* first.

Summary

The five summaries, in brief: the mode identifies the most common category; percentages rather than counts give the fair comparison when groups differ in size; the median reports the typical case while the mean reports totals; the gap between them indicates the shape of the distribution; and the standard deviation measures spread—spread that often reflects mixture rather than chaos. Beneath all five sits the standing precaution to read *N* before trusting anything the statistics appear to show.

Unit II: The Machine — Minnesota's Sentencing System

Module 4: Classification, Felonies, and the Offense Code

Classification, Felonies, and the Offense Code: How the System Decides What a Case

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Felony or misdemeanor? Assault in the first degree or terroristic threats? Criminal justice and law lean heavily on the categorization of human conduct, an activity that is, on close inspection, messy, contested, and rarely as clean as the finished label suggests. The previous unit established a discipline for reading numbers: locate them in the funnel, identify the unit of analysis, and keep description separate from explanation. This chapter turns that discipline on the categories themselves. Before a sentence can be counted, the conduct behind it must be sorted into a class the system recognizes. The Minnesota Offense Code (MOC) that labels each record in the dataset is the last of those sortings. Understanding what the code does, and what it does not, is the aim of this module.

The organizing question is simple: **why do we classify crimes at all?** The answer is that classification is not one act but a chain of them, performed by different actors at successive stages of the criminal-justice funnel, each for a different reason. Following that chain from the top of the funnel to the bottom reveals both the utility of classification and its central limitation, the limitation that governs how the dataset must be read.

Classification Along the Funnel

At the entry point: the police. A crime enters the funnel when police classify conduct as a crime in the first place (*or fails to enter when they do not*), and the felony/misdemeanor line shapes what they may do about it. That line is not merely descriptive; it carries legal force. Under long-settled

Fourth Amendment doctrine, an officer may make a warrantless arrest for a felony on probable cause whether or not the offense occurred in the officer's presence, whereas warrantless arrests for misdemeanors are more tightly constrained: classically, by the requirement that the offense be committed in the officer's presence (LaFave, 2020). Classification, in other words, allocates police authority before it does anything else.

It also allocates attention. Police work is defined less by mechanical rule-following than by discretion, and departments differ systematically in how they exercise it, Wilson's (1968) classic study distinguished "watchman," "legalistic," and "service" styles of policing, each reflecting a different local judgment about which offenses warrant intervention and which do not. The felony line functions here as an administrative threshold: it marks the level of seriousness at which the system commits its more intensive machinery. This is why the felony is, in the draft's phrase, a useful line in the sand. The alternative would be recording and analyzing every speeding ticket and petty theft with the full apparatus reserved for serious crime, a completely untenable situation. Misdemeanors are not trivial in aggregate; they constitute roughly 80 percent of the nation's criminal dockets, some thirteen million cases a year (Natapoff, 2018). But they are handled on a separate, higher-volume, lower-intensity track. The dataset students explore contains none of them. It records only felonies, which are conduct the law designates as most serious, carrying a potential term of more than a year. The Minnesota Sentencing Commission does so precisely because the felony threshold is where the guidelines system trains its attention.

At the charging stage: prosecutors. Once conduct is reported and an arrest made, the prosecutor translates events into a formal accusation. A charging decision is tied to probable cause to believe the accused violated a **specific statute** (Congressional Research Service, 2024), and statutes are themselves acts of categorization: the legislature has already sorted conduct into felony, gross misdemeanor, and misdemeanor bins. But the prosecutor's selection among available charges is

consequential in its own right. Discretion at this stage is broad, and scholars have argued that the charge a prosecutor selects "almost predetermines the outcome of a criminal case," since overcharging and charge bargaining shape which statutory category is ultimately applied (Davis, 1998). This matters for reading the data. As the counting rules from Unit I established, a record is described by its **most serious** offense; lesser charges are absorbed. Charges that drop away through bargaining or merger are a form of attrition **within** categories. This is the next place where events are selected out of the funnel. Put another way this is the same selection logic as the dark figure of crime, those events which never interact with the system at all, only operating one level down. What survives to be coded is not the whole of what happened but the top charge the system chose to pursue.

At the sentencing stage: judges. When a judge imposes a sentence, offense classification becomes a direct input to the outcome. The intellectual foundation is the principle of proportionality. The system intends that the severity of punishment should be proportionate to the seriousness of the offense, this is articulated influentially by von Hirsch (1976), whose "just deserts" framework shaped the development of sentencing guidelines in the United States and, particularly, in Minnesota. Minnesota's guidelines operationalize that principle as a grid whose two axes are offense severity and criminal history. Offense classification is thus one of the grid's two coordinates; it is half of what determines the presumptive sentence recorded in the 'time' column. The mechanics of that grid are the subject of the next module. For now, the point is that classification does not merely describe the case to the judge, it partly *determines* the recommended punishment.

Below and beyond the funnel: the public and the state. A classification does not expire when a sentence is served. Long after the funnel has done its work, the label persists and continues to govern access to ordinary life, a body of restrictions scholars call the *collateral consequences* of

conviction. Chin (2012) describes the aggregate effect as a kind of "civil death": tens of thousands of federal, state, and local provisions restrict the employment, benefits, residence, and civil status of people with convictions, most of whom were never imprisoned at all. Two examples show how finely these consequences track the *type* of felony.

The first is public benefits. A 1996 federal law (the Personal Responsibility and Work Opportunity Reconciliation Act, § 115, codified at 21 U.S.C. § 862a) imposed a lifetime ban on TANF and SNAP, cash assistance and food benefits, for people convicted of a *drug* felony, while permitting states to opt out of or modify the ban. The provision's logic is worth pausing on, because it inverts what one might expect: it applies only to drug felonies and to *no other felony*, however serious. A person convicted of a violent felony, up to and including murder, faces no comparable federal benefits bar; a person convicted of felony drug possession may face a lifetime one. States have largely softened the rule, by late 2023 only one state, South Carolina, had neither opted out of nor modified either ban (Love & Sibilla, 2023), but the example illustrates how a single classificatory line, drawn decades ago for reasons of drug policy, still allocates access to food and cash assistance today.

The second is employment. Employers commonly screen not merely for the *existence* of a felony record but for its *kind*. Research on employer hiring behavior finds that willingness to hire a person with a record varies systematically by offense type: employers are markedly more averse to applicants with violent convictions than to those with drug or property convictions (Holzer, Raphael, & Stoll, 2006). Classification makes that distinction operable—it renders "what kind of felony" a concrete, checkable fact. (Whether such screening is fair, and what harm it produces, is taken up under **Downsides**, below.)

At the far end: scholars. Finally, classification is what makes the systematic study of crime possible at all. It does so in two distinct ways. First, standardization: shared offense definitions let researchers aggregate and compare data across thousands of jurisdictions and across decades, the achievement that national systems such as the FBI's Uniform Crime Reporting program and its successor, the National Incident-Based Reporting System, were built to deliver (Mosher, Miethe, & Hart, 2011). Without common categories, one county's "assault" could not be compared to another's, and no national picture could be assembled. Second, theory-building: offense typologies let scholars generalize beyond individual cases and accumulate knowledge, treating classes of offense as objects of explanation. The construction of crime types is, in this tradition, a precondition for cumulative criminological theory rather than an afterthought to it (Clinard & Quinney, 1973). Every finding in this resource rests on that foundation: the analyses are possible only because the conduct behind 294,000 sentences was sorted into comparable categories in the first place.

The Thesis: The MOC Records a Decision, Not an Event

Trace the chain and its lesson emerges. Police classify conduct as crime; prosecutors translate conduct into a charge; judges translate the charge into a guideline category; and the MOC records the system's final internal label. At each step the category is *re-made*, refined, narrowed, sometimes swapped for another, by human decision. The Minnesota Offense Code is a system used in Minnesota to classify offenses internally, at the level of the state's own recordkeeping. It is not a neutral description of what a person did; it is the residue of a chain of decisions about what the system determined the case to *be*.

This is Sellin's Dictum from Unit I, applied now to categories rather than to counts. Just as felony sentencing data is an excellent record of what the system *decided* and a poor proxy for how much

crime *occurred*, the MOC is an excellent record of the system's final classification and a poor proxy for the underlying conduct. A record coded to a given offense reflects the charge that survived every prior stage, not the full reality of the event. Reading the code as a transparent window onto behavior is the categorical version of the same error Unit I warned against. This beckons further discussion, but entire entries are only representative of the *highest*, most severe count, and in a case where someone murders one person and nearly kills another only the murder would appear as an entry, thus the dataset is not indicative of the question "How many people were attacked?"

The Four Families

Beneath the fine grain of the MOC lies a coarse partition that criminologists use to organize almost any body of offense data. Conventionally, offenses are grouped into four "families": **person** (harm directed at a body), **property** (harm to ownership or possession), **drug** (offenses defined by controlled substances), and **public order** (offenses against collective norms or the functioning of institutions). This four-way scheme is the working vocabulary of much criminal-justice statistics. The Bureau of Justice Statistics, for instance, routinely reports violent, property, drug, and public-order offenses as distinct groups. It maps closely, though not exactly, onto the official federal taxonomy: NIBRS sorts every offense into **Crimes Against Persons**, **Crimes Against Property**, and **Crimes Against Society**, folding both drug and public-order offenses into that third "Society" category (Federal Bureau of Investigation, n.d.). The module's four families simply split "Crimes Against Society" back into its drug and public-order halves.

That the official scheme has three categories and the analytic one has four is a first clue to a deeper point: **categorization is an act of human judgment, not a reading of nature**. Boundary cases make this vivid. Robbery, theft accomplished by force, sits astride the line between the person and property families, and different systems resolve it differently: it is a *person* crime under

the FBI's older Summary Reporting System but a *property* crime under NIBRS. Nothing about robbery changed; the classifiers made a different choice. Such edge cases are features, not flaws. They reveal the work a classification is doing.

The most influential account of that work comes from outside criminology. In **Sorting Things Out**, Bowker and Star (1999) argue that "to classify is human" (p. 1), that every classification system, however technical it appears, encodes choices and values, advantaging some perspectives while silencing others, and that this constructedness is usually rendered invisible by the finished categories themselves. Their lesson reprises the discipline from Module 1, where every column was to be treated as an administrative decision first and a measurement second. The MOC is exactly such a decision. It was built to run Minnesota's sentencing system, and its categories carry the priorities of that system rather than the neutral contours of human conduct.

Working with the MOC in the Explorer, one navigates this classification by *drilling down* one character at a time. The first character, ``moc1``, names the top-level family; successive characters narrow the offense within it. The exercises below explore one top-level code exemplifying each family, to see how each type of offense is classified and, more instructively, to see where the categories surprise us.

> **Explorer Task A.** Read the value counts for ``moc1`` on the full dataset. Note how large a share the person-crime categories that dominate public attention: homicide & assault, actually occupy among *felonies*, the most serious offenses of all.

> **Explorer Task B.** Drill into assault (``moc1 == A``) and read the sub-code distribution. Observe what actually dominates the category, whether physical assault, or the verbal-threat offenses one might not expect to find at the top of an "assault" bin.

The second task previews a recurring theme: what a category *contains* is often not what its name implies. That gap between label and content is not a defect in the data but a fact about how classification and charging work and it is one reason media impressions of crime mislead.

Why the small-share finding is surprising. Homicide and violent assault command a share of public attention wildly out of proportion to their frequency, because news coverage overrepresents them. A content analysis of a full year of local crime reporting in Milwaukee found that homicide accounted for a majority of the front-page crime stories in the daily newspaper and a large share online, even though homicides are so rare relative to total crime as to be nearly invisible in the underlying police statistics; property offenses, conversely, made up the great bulk of actual crime but a small fraction of coverage (O'Hear, 2020). The pattern is long-standing. Decades earlier, researchers found that the public's picture of crime tracked newspaper portrayals far more closely than official crime counts (Sheley & Ashkins, 1981). This is why discovering the modest share of violent offenses even among felonies can feel counterintuitive: the data corrects an impression the media has built. What the dataset's modal categories reveal is not what crime *feels* like but what the criminal legal system actually processes.

Downsides to Classification

These classifications are indispensable, as the walk down the funnel showed. They are also open to serious criticism.

The first criticism concerns *coherence*. Classification is meant to track relative seriousness. In premise it aims to sort conduct so that graver wrongs sit in graver categories. Scholars have shown that the felony/gross-misdemeanor/misdemeanor grading of statutes often drifts from that ideal. In an empirical study, a research team surveyed a representative sample of Pennsylvanians about the seriousness of many offenses and compared public judgment to the statutory grades those

offenses actually carried, finding frequent and substantial mismatches (Robinson, Gaeta, Majarian, Schultz, & Weck, 2010). In one striking instance, possessing wiretapping equipment was rated by the public as roughly comparable to stealing goods worth \$50 to \$200, an offense whose grade implies a maximum of about two years, yet the wiretapping offense is graded as a felony carrying up to seven years. The difference is not merely academic; the grade determines exposure to years of imprisonment. If grading can diverge this far from shared intuitions of seriousness, then the categories students analyze are administrative artifacts to be read critically, not natural measures of harm.

The second criticism concerns *consequences*. The walk down the funnel noted that classifications outlive the sentence, governing benefits and employment. Those uses carry real benefits for employers and agencies, but they also inflict real harm. The foundational evidence comes from an audit study in which matched applicants, differing only in whether they reported a criminal record, applied for entry-level jobs: a felony record roughly halved the callback rate for white applicants and cut it by nearly two-thirds for Black applicants (Pager, 2003). It is also worth considering however that later field experiments confirm the record's weight but complicate the remedies: when employers were barred from asking about records at all, some appeared to substitute racial assumptions, widening the very gap the policy sought to close (Agan & Starr, 2018). Classification, then, is double-edged. The same categorical precision that lets the state monitor its own sentencing also lets a label follow a person for a lifetime.

Summary

Classification is the machinery that turns human conduct into a countable record, and it operates at every stage of the funnel. Police use the felony line to allocate authority and attention and to set the threshold for intensive recordkeeping; prosecutors translate conduct into charges, selecting

among categories in ways that shape the outcome; judges use offense severity as one of the two axes that fix the presumptive sentence; the public and the state use the surviving label to govern benefits and employment long after the case closes; and scholars depend on standardized categories both to compare data and to build theory. The Minnesota Offense Code sits at the end of this chain. It records what the system decided a case **was**, not what happened—Sellin's dictum in categorical form. Beneath its fine grain lie four families—person, property, drug, and public order—whose boundaries are acts of human judgment rather than facts of nature, as edge cases like robbery and the mismatches of statutory grading both reveal. Read this way, the MOC is neither a neutral mirror of crime nor a flaw to be corrected, but an administrative decision to be interpreted with care. The next module opens the grid on which offense severity and criminal history combine to produce the number in the `time` column.

References

- Agan, A., & Starr, S. (2018). Ban the box, criminal records, and racial discrimination: A field experiment. **The Quarterly Journal of Economics, 133*(1), 191–235.*
<https://doi.org/10.1093/qje/qjx028>
- Bowker, G. C., & Star, S. L. (1999). **Sorting things out: Classification and its consequences.** MIT Press.
- Bureau of Justice Statistics. (n.d.). **Data collections and statistical reports.** U.S. Department of Justice. <https://bjs.ojp.gov/> **[Cite a specific dated BJS report using the violent/property/drug/public-order scheme, with NCJ number, before publication.]**
- Chin, G. J. (2012). The new civil death: Rethinking punishment in the era of mass conviction. **University of Pennsylvania Law Review, 160*(6), 1789–1833.*

Clinard, M. B., & Quinney, R. (1973). **Criminal behavior systems: A typology** (2nd ed.). Holt, Rinehart and Winston.

Congressional Research Service. (2024). **Federal prosecutorial discretion: A brief overview** (LSB11326). Library of Congress. <https://www.congress.gov/crs-product/LSB11326>

Davis, A. J. (1998). Prosecution and race: The power and privilege of discretion. **Fordham Law Review*, 67*(1), 13–67.

Federal Bureau of Investigation. (n.d.). **A guide to understanding NIBRS.** U.S. Department of Justice. <https://ucr.fbi.gov/nibrs/2011/resources/a-guide-to-understanding-nibrs>

Holzer, H. J., Raphael, S., & Stoll, M. A. (2006). Perceived criminality, criminal background checks, and the racial hiring practices of employers. **The Journal of Law and Economics*, 49*(2), 451–480. <https://doi.org/10.1086/501089>

LaFave, W. R. (2020). **Search and seizure: A treatise on the Fourth Amendment** (6th ed.). Thomson Reuters.

Love, M., & Sibilla, N. (2023). **Accessing SNAP and TANF benefits after a drug conviction: A survey of state laws.** Collateral Consequences Resource Center.

Mosher, C. J., Miethe, T. D., & Hart, T. C. (2011). **The mismeasure of crime** (2nd ed.). SAGE Publications.

Natapoff, A. (2018). **Punishment without crime: How our massive misdemeanor system traps the innocent and makes America more unequal.** Basic Books.

O'Hear, M. M. (2020). Violent crime and media coverage in one city: A statistical snapshot.

**Marquette Law Review, 103*(3), 1007–1034.*

Pager, D. (2003). The mark of a criminal record. **American Journal of Sociology, 108*(5), 937–975.*

<https://doi.org/10.1086/374403>

Personal Responsibility and Work Opportunity Reconciliation Act of 1996, Pub. L. No. 104-193, § 115, 21 U.S.C. § 862a.

Robinson, P. H., Gaeta, T., Majarian, M., Schultz, M., & Weck, D. (2010). The modern irrationalities of American criminal codes: An empirical study of offense grading. **Journal of Criminal Law and Criminology, 100*(3), 709–764.*

Sheley, J. F., & Ashkins, C. D. (1981). Crime, crime news, and crime views. **Public Opinion Quarterly, 45*(4), 492–506.* <https://doi.org/10.1086/268684>

von Hirsch, A. (1976). **Doing justice: The choice of punishments.** Hill and Wang.

Wilson, J. Q. (1968). **Varieties of police behavior: The management of law and order in eight communities.** Harvard University Press.

Module 5: The Guidelines Grid

Minnesota as the nation's first presumptive-guidelines state (1980); the grid: **offense

severity × criminal-history score → presumptive sentence**; `time` as the grid's output.

Reprise variance-as-composition at full strength: inside one grid cell, the presumptive

sentence barely varies — **the grid is visible in the std**. Lab: reconstruct two grid cells

by filtering and compare their summaries.

Assumes working severity-level column; `history` confirmed. If severity is absent, refit around `history` × offense family.*

Module 6: Criminal History — The Second Axis

How the history score accumulates (prior felonies, custody status, decay rules — kept conceptual); the distribution of `history` in the data (heavily bottom-weighted); how median `time` climbs with `history` **descriptively**. First rehearsal of precise language: "sentences are longer at higher history scores" vs. "history causes longer sentences" — here the causal claim is actually **by design** (the grid encodes it), which makes this the safe place to teach the difference between a pattern and a mechanism.

Data anchor: median `time` by `history` level, and the share of counts at `history == 0`.

Module 7: What `time` Measures — Presumptive vs. Pronounced

Throughout these materials you've been using the variable 'time' extensively, but until now we haven't paused to consider what 'time' represents. A cursory assumption would be that it represents how many years a sentenced felon is going to serve, but the reality is far more complicated.

What the number 'time' truly represents is the presumptive sentence in number of months. In reality very few inmates serve out their full sentence in Minnesota (cite percentage), instead there are mechanisms within the system which reduce that number over time. (dive into those systems)

The gap between the guideline recommendation and what happens: stayed vs. executed sentences,

dispositional and durational ****departures****, plea bargaining's shadow. Lab: audit how stayed / probationary outcomes are coded in `time` (the ZERO-CHECK from Module-3 development) and, if

departure/disposition columns exist, describe departure rates by offense family.

Hard dependency on disposition/departure columns. If absent, this module contracts to a one-week "coding audit + limitations" chapter — which is still genuinely valuable.*

Unit III — Reading Patterns Responsibly

Module 8: Change Over Time

`sentyear` as a variable; volume and median-sentence trends 2001–2019; anchoring trends to policy events — centrally the ****2016 Minnesota drug sentencing reform**** (verify effective dates and which offenses; the dataset holds ~3 post-reform years). The lesson in method: a discontinuity that lines up with a law is *suggestive**, not proof — composition changes too.

Data anchor to hunt: drug-offense median `time` and volume before/after 2016.

Platform prerequisite: this is the module where the absence of line charts hurts most — schedule after visualization ships, or lean on small year-by-year tables.

Module 9: Holding Things Constant

The course's methodological hinge. Comparing groups fairly: composition effects, the filter-as-control move ("how much of the pattern survives when I hold this constant?"), and — if the data yields one — a real ****Simpson's-paradox**** example where a gap reverses inside subgroups. Lab: take a raw gap, add one control filter at a time, narrate what happens to the gap at each step.

Data anchor to hunt: a genuine sign-reversal or major attenuation example; offense-mix differences between two groups are the most likely source.

Module 10: Small Numbers, Big Mistakes

Expansion of check-N-first: rare offenses, single-county single-year cells, the volatility of small-cell means across adjacent years (same filter, consecutive years, wildly different means — nothing changed but noise). The fireworks three return as old friends.

Data anchor: one rare offense whose year-to-year mean swings absurdly.

Module 11: Sex and Sentencing

The gender gap as the **methodological dress rehearsal** for Unit IV: women are rarer in the system and sentenced more shortly — now apply Module 9's toolkit. Does the gap survive

holding offense family constant? History score? What remains, and what **can't** this data say about why? The pattern is among the most robust in the sentencing literature, which makes it the right place to practice disparity reasoning with lower interpretive stakes.

Builds directly on the Module-3 base-rate example.

Learning objectives. By the end, students can (1) distinguish a raw gap from a conditional gap; (2) execute a stratified comparison in the Explorer using offense family and criminal-history score; (3) articulate what a residual gap does and does not establish; (4) name at least three candidate explanations the dataset cannot adjudicate between.

1. Opening hook: two numbers that can't both be the headline (~3 min)

Present the raw pattern without commentary: women are roughly 15–20% of sentencing records, and their mean pronounced time is substantially shorter than men's. Pose the framing question students should now anticipate: *is this one fact or two?* Fewer women in the data and shorter sentences for women are separate observations, sitting at different funnel depths. Explicit callback to Module 3: recall that men will out-count women on nearly everything simply because of group size — so every comparison in this module must be in percentages and group-wise means, never raw counts.

2. Why gender first, race later (~2 min)

Name the pedagogical move openly. The gender gap is among the most consistently replicated findings in the sentencing literature (Daly & Bordt; Starr's federal work; MSGC's own monitoring reports), which makes it a low-stakes arena to practice the reasoning: students can focus on *method* without the interpretive weight Unit IV will carry. State the promise: every move made in this module will be repeated there, with the same discipline.

3. The naive comparison and why it's not enough (~4 min)

Run the unconditioned crosstab: sex × mean time. Establish the gap's size. Then rehearse the objection students should raise themselves by now: men and women may not be *in* the data for the same offenses. If women's records cluster in lower-severity offense families (property, drug) and men's in person offenses, the raw gap partly reflects *composition*, not *treatment*. This is Simpson's-paradox territory — flag the name, keep the math informal.

4. First conditioning: hold offense family constant (~5 min)

Drill into moc1 families one at a time and re-run sex × mean time within each. Two findings to surface: (a) the composition story is partly right — the mix differs; (b) the gap *shrinks but does not vanish* within most families. Introduce the vocabulary formally: the *within-stratum* gap, the *residual* gap.

5. Second conditioning: hold criminal history constant (~5 min)

Repeat within history bands (0, 1–2, 3+), ideally within a single offense family so students see two-way conditioning. Key observation: women's history scores skew lower, so some of the raw gap flows through history — which is itself an interesting fact, since history scores encode *prior* system contact, meaning earlier funnel stages are baked into this variable. The gap narrows again. Something remains.

6. The hard part: what the residual gap can't tell you (~5 min)

The module's payload. The residual gap is a *description that has survived two adjustments* — it is still not an explanation. Walk the candidate explanations the data cannot separate:

Judicial leniency (differential treatment at sentencing)

Charging and plea decisions upstream — the funnel again; identical conduct may enter the data as different offense codes by sex

Unrecorded legally relevant factors — role in the offense, weapon use, victim injury within the same moc code

Unrecorded extralegal factors — caregiving responsibilities, which courts may weigh through departures (Minnesota's dispositional departures are a natural pointer here; MSGC reports show women receive mitigated dispositional departures at higher rates)

Selection into the dataset itself — if marginal female cases are diverted pre-felony, the women who remain are non-random

Close with the discipline restated: the Explorer can shrink a gap; it cannot assign a cause. Adjudicating between these requires data the monitoring system was never designed to collect (Module 1's found-data lesson, returned).

7. Checkpoint + bridge to Unit IV (~2–3 min)

Short quiz (see tasks below), then the bridge: name the exact sequence just performed — raw gap → composition check → conditioning on offense → conditioning on history → residual gap → enumerate unobservables — and state that Unit IV applies this sequence verbatim, where the stakes and the temptation to skip steps are both higher.

Interactive task ideas within the Explorer

Core sequence (in-module)

Reproduce the raw gap. Clear history, run column summary on sex (counts and percentages), then crosstab sex × mean time. Record both numbers — reinforces that "rarer" and "shorter" are two findings.

Composition audit. Crosstab sex × moc1 in percentages. Question: which offense family shows the largest male–female difference in *share of that group's records*? (Base-rate reasoning applied.)

One-family conditioning. Filter to drug offenses; re-run sex × mean time. Quiz: did the gap grow, shrink, or hold? Repeat for person offenses; compare.

Two-way conditioning. Within one offense family, filter history == 0 and compare sex means; then history >= 3. Question: is the gap larger among first-timers or repeat offenders — and write one sentence on why that's interesting *without* claiming to know the cause.

The trap task. Ask students to find a stratum where the gap *reverses* or nearly vanishes, then explain why one cell doesn't overturn the overall pattern (small-n cells, mode/mean instability — Module 3 callback).

"What can't this view say?" Free-response: given the final conditioned view, list two explanations consistent with it that the dataset cannot distinguish. Graded on enumerating unobservables, not on picking a favorite.

Extension / optional

Median vs. mean check. Re-run a key comparison with medians. If the gap changes size, connect to skew and the mean–median lesson from Module 3.

Time trend. Crosstab sentyear × mean time split by sex: has the gap narrowed since 2001? Caution prompt: guideline changes over time are a confound.

Severity-weight probe. Compare twweight distributions by sex within an offense family — a proxy for whether men's cases carry more additional counts, i.e., composition hiding *inside* a stratum.

Predict-then-check. Before running task 3, students write a predicted direction and magnitude; the Explorer adjudicates. Builds the habit of registering expectations before looking.

Simpson's construction challenge. Advanced: find (or approximate) a pair of strata where within-group gaps point one way but the pooled gap points differently, and explain the aggregation mechanics.

Write the honest headline. Students draft a one-sentence, publication-ready summary of the module's finding that a methodologist couldn't object to — practicing the descriptive register ("women's sentences average X months shorter within the same offense family and history band") over the causal one ("courts favor women").

Unit IV — The Big Question

Module 12: Race and Sentencing Disparity

The course's reason for its own caution. ****Disparity vs. discrimination**** as the load-bearing distinction; the naive `race` crosstab, then the Module-9 procedure applied with full care: offense family, severity, history — watching the gap change and being precise about what the residual does and does not establish (unobserved confounders, upstream charging and plea decisions, selection into the dataset). Descriptive data can **document** disparity; it cannot, alone, adjudicate discrimination.

*Prerequisites: Modules 9 and 11 completed; educator teaching-notes feature shipped

(EDUCATOR_PORTAL.md, feature 12); every claim in the narration verified against the data.*

Module 13: What the Data Can't Tell You — and Who Can

The honest ceiling. How researchers go beyond description (conceptual-only tour: regression, matching, the idea of "controlling for" done formally); how to read a sentencing-disparity claim in the news and locate its evidence type; revisiting the funnel — every analysis in this course is conditional on who reached sentencing. Closes the interpretive arc opened in

Module 1.

Unit V — Capstone (Weeks 14–15)

Module 14: Designing a Descriptive Investigation

Students pose their own answerable question, pre-specify a filter plan, run it, and check N at every node. The platform's deep feature becomes the pedagogy: ***the filter history *is* the methods section*** — reproducible by anyone who replays the chain.

This one could be simple but deceptively difficult to present to an educator to grade.

Module 15: Communicating Findings

Writing with statistics: choosing mean vs. median and defending the choice; uncertainty language; precise descriptive claims (no causal leakage); a short written report whose appendix is the filter-token chain. Peer review against a rubric drawn from Modules 1, 9, 10, and 12.